Hw 8

3. $x \cdot [2,3] + y \cdot [1,-2] = [6,16]$

[2x2] linear system of equ.

$2x + y = 6$
$3x - 2y = 16$

$\to y = 6 - 2x \to y = 6 - 2(4) = -2$

$3x - 2(6-2x) = 16$

$3x - 12 + 4x = 16 \to 7x = 28,\ x = 4$

$\boxed{x = 4,\ y = -2}$

Verification:

$4 \cdot [2,3] + (-2) \cdot [1,-2] = [8,12] + [-2,4] = [6,16]$

Hw 8 Note:

Want: $a \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix} + c \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ 11 \end{bmatrix}$ Target vector

Matrix equation : $Ax = b$ These 2 are equivalent

$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 3 \\ -1 & 4 & -5 \end{bmatrix}$, $x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ -9 \\ 11 \end{bmatrix}$